

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

1st Semester of B. Pharm. Examination (New)

University Theory Examination April 2016

PH112 – Human Anatomy and Physiology - I

Date: 13/04/2016, Wednesday Time: 10.00 a.m. to 1.00 p.m. Maximum Marks: 80

Instructions:

1. There are three sections in this question paper.
2. **SECTION – I** comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to **SECTION – I** are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for **SECTION – I** is 30 minutes. Answers to **SECTION – I** must be written during the first 30 minutes of the examination.
3. Answers to **SECTION – II** and **SECTION – III** are to be provided in separate Main Answer Books provided to you.
4. Figures to right indicate marks.
5. Draw neat sketches wherever necessary.

SECTION - I

Q.1 Attempt all questions. Each question is of one mark.

20

1. Most of the glucose that is filtered through the glomerulus undergoes reabsorption in the
 - A. descending limb of the loop of Henle
 - B. ascending limb of the loop of Henle
 - C. distal tubule
 - D. proximal tubule
2. In RBC lifecycle Iron is removed from heme portion in form of Fe^{3+} through which transporter?
 - A. Erythropoietin
 - B. Transferrin
 - C. Ferritin
 - D. Globin
3. Which type of WBC cells' Nucleus have a shape of Kidney?
 - A. Monocyte
 - B. Basophils
 - C. Neutrophil
 - D. Lymphocyte
4. Which of the following is not a part of plasma protein?
 - A. Albumin
 - B. Fibrinogen
 - C. Bilirubin
 - D. Globulin
5. Which of the following considered as Natural Killer cells in immune response?
 - A. Lymphocyte
 - B. Monocyte
 - C. Eosinophils
 - D. Basophils

6. A basic mechanism involved in concentrating the urine is
- Diffusion
 - Countercurrent multiplier system
 - Active transport
 - Passive transport
7. Choose the correct sequence of current flow through the heart wall.
- AV node, SA node, Purkinje fibers, AV bundle of His, right and left bundle branches
 - SA node, AV node, AV bundle of His, right and left bundle branches, Purkinje fibers
 - SA node, Purkinje fibers, AV node, AV bundle of His, right and left bundle branches
 - AV node, Purkinje fibers, AV node, AV bundle of His, right and left bundle branches
8. During pulmonary circulation blood leaves the
- right ventricle and goes directly to the aorta.
 - right atrium and goes directly to the lungs.
 - right ventricle and moves to the lungs.
 - right atrium and goes directly to the left ventricle.
9. Pyelonephritis is
- dislocation of kidney
 - tumour of kidney
 - inflammation of kidney
 - infection of kidney
10. The walls of the alveoli are composed of two types of cells, type I and type II. The function of type II is to _____
- trap dust and other debris
 - secrete surfactant
 - replace mucus in the alveoli
 - protect the lungs from bacterial invasion
11. The region where a bronchus and pulmonary vessel enter and exit the lung is
- the capsule
 - the base
 - the hilum
 - the apex
12. Which of the following occurs during the chloride shift?
- Bicarbonate is removed from the blood
 - Chloride is removed from the erythrocyte
 - Bicarbonate is removed from the erythrocyte
 - Chloride is exchanged for bicarbonate
13. Which one of this is the 12th factor in clotting process?
- Fibrin stabilizing factor
 - Fibrinogen
 - Hageman factor
 - Christmas factor
14. Which chemical is a strong inhibitor of Platelet adhesion & release?
- Heparin
 - Prostacyclin
 - TXA₂
 - Plasminogen

15. All of the following are accessory digestive organs except _____.
 - A. the teeth
 - B. salivary glands
 - C. the liver
 - D. the spleen
16. What is the full form of RAAS?
 - A. Renin angiotensin aldosterone system
 - B. Renin adrenergic system
 - C. Renin alkaline aldosterone system
 - D. Reticulo angiotensin aldosterone system
17. Which of the following is not a function of the stomach?
 - A. formation of chime
 - B. secretion of gastric juice
 - C. absorption of fats
 - D. storage of ingested food
18. One of the following is not a disease / condition of Respiratory system.
 - A. Asthma
 - B. Bronchitis
 - C. Cystitis
 - D. COPD
19. The exocrine secretory units of the pancreas are called _____.
 - A. Beta cells
 - B. Islet cells
 - C. Acini
 - D. Alpha cells
20. The function of G-cells is to _____.
 - A. secrete gastrin hormone
 - B. secrete gastric acid
 - C. secrete gastric lipase
 - D. secrete somatostatin

SECTION – II

Q 2 Attempt any FOUR of the following

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|---|--|----|
| A | Explain a brief physiology of Heart & enlist major disorders of cardiovascular system | 05 |
| B | Explain in brief about blood circulation in cardiovascular system | 05 |
| C | Provide detail anatomy of the heart with diagram | 05 |
| D | Explain concept of cardiac cycle in detail | 05 |
| E | Explain regulation of Blood pressure in body and factors affecting the same | 05 |
| F | Give following definitions:
ECG, Heart rate, Blood pressure, Cardiac out put, heart sound | 05 |

SECTION – III

Q 3 Attempt any FOUR of the following

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|---|---|----|
| A | Describe the composition and function of blood | 05 |
| B | Define respiration. Explain in detail mechanism of internal respiration | 05 |
| C | Enumerate various clotting factors and explain in detail mechanism of blood coagulation | 05 |
| D | Draw a labeled diagram of lung and describe its function. | 05 |
| E | Write a note on blood group and its significance | 05 |
| F | Write a note on respiratory volumes | 05 |

Q 4 Attempt any FOUR of the following

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|---|--|----|
| A | Explain the physiology of urine formation | 05 |
| B | Draw a neat and labeled diagram of nephron | 05 |
| C | Draw a neat labeled diagram of digestive system and enumerate functions of each of the labeled parts | 05 |
| D | Describe composition and function of gastric juice | 05 |
| E | Write a note on disorders of urinary system | 05 |
| F | Draw a neat labeled diagram of liver and describe the structure and function of liver | 05 |
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